# Chapter 21

# Specifications

Understanding specifications and tolerances is important when applying measurement theory to the real world. The terms *specification* and *tolerance* are closely related and often confused. Although both terms define quantitative limits in relation to a nominal value, they are used in different situations. Unfortunately, some definitions even conflict with one another for the same term.

*Specification* defines the limits within which an instrument is able to be useful. Properly interpreted and used, specifications are useful in evaluating the performance of inspection, measurement, and test equipment (IM&TE) during calibration and for evaluating the capability of a measurement standard to perform a calibration.

*Tolerance* is the maximum permissible variations or limits allowed a quantity/parameter from a specified value. Information is described in the form of "$\pm$ n %," "not less than *n* dB," "maximum," "minimum," etc., in the performance requirement section of the calibration procedures of measuring and test equipment and measurement standards being calibrated. Tolerances are further limits *within* specifications.

Here are other sources of definitions of the words *specification* and *tolerance*:

- *Specifications* define the expected performance limits of a large group of substantially identical finished products (all units of a specific model of digital thermometer, for example). Customers use specifications to determine the suitability of a product for their own applications.
- In JCGM 200-2012, *VIM* 3, *specification* is referenced in the first note for "measurand," where measurand is "quantity intended to be measured."
    - "NOTE 1 The **specification** of a measurand requires knowledge of the *kind of quantity*, description of the state of the phenomenon, body, or substance carrying the quantity, including any relevant component, and the chemical entities involved."[1]
- *VIM* 3 also mentions "specification" in **maximum permissible measurement error** (maximum permissible error, limit of error) (*VIM* 3, 4.26): "extreme value of *measurement error,* with respect to a known *reference quantity value*, permitted by **specification or regulations** for a given *measurement, measuring instrument,* or *measuring system*."[2]

## Application of Specification and Tolerance

The essential result of a calibration (ignoring the data for now) is a pass/fail decision about the item. The decision is based on the results of one or more measurements. Historically, the terminology has been that the result of a measurement is *in tolerance* or *out of tolerance*. This is probably based on the sense that calibration is a process that acts on one item at a time, so the result applies to only one item. It should be understood, however, that the performance specifications of an instrument or tool are the basis of a calibration procedure, as they are what the performance is evaluated against. It is more correct to say that the result of a measurement is within specification or out of specification.

# TYPES OF SPECIFICATION LIMITS

There are two general types of specification limits. Which is used depends on the equipment, application, and what the information is used for.

1. A *one-way specification* allows variation in only one direction from the nominal value. The result of a measurement may be either more or less than one of the limits, but not both.[12] The nominal value is either a lower limit or an upper limit of the specification. This type of limit is often found in instruments used for work related to safety. Some companies may refer to a one-way specification as *unilateral* or *single-sided*.[13]

   Here are two examples of one-way specification:

   - An insulation tester may have a specification that the applied test voltage is 500 V, –0 % + 10 % when the test voltage selector is in the 500 V position. The purpose of this is to ensure that at least the specified test voltage is applied to the tested part to ensure that the part meets safety requirements. For the calibration technician, without considering the impact of measurement uncertainty, any measured value from 500 V to 550 V would pass, but 499.9 and 550.1 would both fail.
   - An appliance safety tester may specify that an alarm is to sound if a leakage current is 1 mA, –25 % + 0 %. The purpose of this test is to ensure that the user of the appliance cannot receive a dangerous shock. In this case, the instrument would fail if the applied current exceeded 1 mA and the alarm still had not sounded.

2. A *two-way specification* allows variation in either direction from the nominal value. The nominal value is a target, and the specification defines lower and upper limits of acceptable values.[14] The nominal value is not necessarily centered between the limits. Some companies may refer to a two-way specification as *bilateral*, or as *double-sided*.[15]

   Here are three examples of a two-way specification:

   - The output of a voltage source may have a specification of ± (0.005 % of setting + 10 mV). If the calibration technician tests it at 500 mV, the specification limits will be 499.965 mV to 500.035 mV.